CSE-56

PROJECT ABSTRACT [Form – 1] SESSION (2026-27)

PROJECT ID: SKIT/ CSE /2023-2027/56 BRANCH: Computer Science & Engineering **SECTION:** D

TITLE OF PROJECT: Automated Attendence System for Rural Schools

SDG MAPPING: Quality Education

PROJECT TRACK:

R&D INNOVATION	CONSULTANCY PROJECTS	DEPARTMENT AS CLIENT PROJECTS	CLIENT PROJECT ✓	HARDWARE-BASED PROJECTS

MANDATORY EXTERNAL EVALUATION OF PROJECT (Tick the appropriate one)

RESEARCH PAPER PUBLICATION ✓	PATENT FILING / GRANT	POSTER COMPETITION

PROBLEM STATEMENT:

To develop a software only, web-based facial-recognition attendence System that works in a phone browser, identifies students automatically using a camera, continues functioning when internet connectivity is limited, and synchronises attendence records to a central database once connectivity becomes available - reducing dependence on manual, paper-based registers in rural schools.

TECHNOLOGY STACK:

NAME OF TOOL / TECHNOLOGY	FRONTEND /BACKEND	SOFTWARE / HARDWARE	PURPOSE OF USE
HTML, CSS, Javascript	frontend	Software	Responsive teacher-friendly phone-browser interface.
flask	Backend	Software	Web backend, dashboard and API integration.
Open CV	Backend	Software	face detection and image processing.
Pretrained CNN face-Recognition Model	Backend/ AI	Software	student identity verification.
SQLite	Backend/ Database	Software	local, on-device attendence storage for offline use.

PostgreSQL	Backend / Database	Software	Central cloud storage of student profiles and attendence records, synced when online.
Render / Railway	Backend / Deployment	Software	Proposed hosting environment for the cloud Database.

PROPOSED PROJECT SPRINTS (add more row if required):

S.No.	SPRINT NAME	PROPOSED USER STORY OF RESPECTIVE SPRINT
1.	To data collection & Model training	To collect student face-image data and build the initial face-detection / recognition prototype.
2.	Model Testing & Integration	To test the recognition model's accuracy and connect it with the application flow.
3.	Backend & Local Database	To build the flask backend and set up local / cloud database storage for attendance data.
4.	UI design & API linking	To build a mobile friendly frontend and connect it with backend APIs.
5.	offline sychronization	To test offline functionality and data synchronization between local and cloud databases.
6.	Pilot Deployment	To test the complete system end to end at a partner school and fix identified bugs.

TEAM MEMBER DETAILS:

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Vageesha Choudhary	8070088880	Backend Development, AI/ML, Project Management	Team Lead & AI/ML Engineer: face detection / recognition prototype, model testing & integration, flask backend, overall coordination.

Shivani Paujdar	8800794325	PostgreSQL, Database Design	Database developer – facial data schema design, Postgres setup, offline-sync testing data integrity verification.
Shivanshi Shringi	8209576234	HTML, CSS, Javascript, UI development.	frontend developer – enrollment & recognition- result screens, teacher dashboard, mobile UI testing, final pilot presentation.

Verified & Approved By:

Mentor's Name & Signature

Pawan Kr. Patidar

22/8/26

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form – 2]

PROJECT ID: SKIT/AI/2023-2027/56 BRANCH: ~~Computer science & Engineering~~ **SECTION:** D

TITLE OF PROJECT: Automated Attendance System for Rural Schools.

TEAM LEAD: Vageesha choudhary		**SPRINT(s) NAME :** Data preparation & Model Prototyping Model Testing & Integration, Backend & Local Database, UI design & API linking, Testing & offline synchronization, Pilot Deployment & finalization.	
USER STORY	**START DATE**	**END DATE**	**DETAILS OF TASK TO BE COMPLETED UNDER THE USER STORY**
To collect face-image dataset & build face-detection recognition prototype.	10 Aug 2026	3 Oct 2026	To be collected around 500 student face images across varied lighting and angles; set up the OpenCV face-detection pipeline; began integrating the pretrained CNN model to build the initial face-detection / recognition prototype.
Tune recognition model & integrate into attendance flow	4 Oct 2026	31 Oct 2026	To be fine-tuned the recognition confidence threshold to balance false-accepts / rejects; benchmarked recognition speed on low-end Android phones; integrated the recognition model into the app's attendance-marking flow.
To Build flask backend APIs & integrate database layers	1 NOV 2026	2 Jan 2027	To be Built flask REST API routes for enroll, recognize, sync and attendence-report; wired the API to the recognition module. Coordinated local (SQLite) and Cloud database integration; set up teacher-account authentication.

To connect backend APIs to frontend & optimize performance.	3 Jan 2027	6 Feb 2027	To be connected the recognition and attendence API endpoints to the frontend; optimized API response times; added logging for failed sync attempts.
To test offline sync & lead pilot deployment	7 Feb 2027	6 Mar 2027	To be coordinated end to end offline-mode testing - recognized and stored attendance with no internet, then verified auto-sync on reconnect; led the onsite pilot deployment at the partner rural school and oversaw final system testing and project submission.

MEMBER 1 : Shivani Faujdar — **SPRINT(s) NAME :** Data preparation & Model Testing & Integration, Backend & local Database, UI Design & API linking, Testing & Offline synchronization Pilot Deployment & finalization.

USER STORY	START DATE	END DATE	DETAILS OF TASK TO BE COMPLETED UNDER THE USER STORY
To Design database schema & begin labelling face dataset.	10 Aug 2026	3 Oct 2026	To be Designed the student profile and facial embedding database schema (SQLite local + PostgreSQL cloud tables); document field mappings for name, roll no., class and embedding vector, and began labelling the collected face dataset.
To Build attendance log table & validate recognition data	4 Oct 2026	31 Oct 2026	To be validated recognition results against seeded student records; built the attendance-log table with timestamp, student-ID and confidence-score fields; stress-tested SQLite writes under repeated recognition calls.
To setup PostgreSQL cloud database & write sync script	1 Nov 2026	2 Jan 2027	To be setup the PostgreSQL cloud database on Render / Railway with student profile and attendance tables; wrote the sync script to push local SQLite records to PostgreSQL once online; added conflict handling for duplicate records.
To link dashboard queries to database & build CSV export.	3 Jan 2027	6 Feb 2027	To be linked the teacher-dashboard queries to PostgreSQL for attendance reports; built the export-to-CSV feature for monthly attendance summaries.

To test database sync accuracy & verify records.	7 Feb 2027	6 Mar 2027	To be Tested sync accuracy and data integrity between local SQLite and cloud PostgreSQL under intermittent connectivity; performed final database verification and record-integrity checks after the pilot.

MEMBER 2 : Shivanshi Shringi

SPRINT(s) NAME : Data preparation & Model Testing & integration, Backend & Local Database, UI Design & API linking, Testing & offline synchronization, Pilot Deployment & Finalization.

USER STORY	START DATE	END DATE	DETAILS OF TASK TO BE COMPLETED UNDER THE USER STORY
To Build Student enrollment & face-capture interface	1 Aug 2026	3 Oct 2026	To Build the student enrollment form for capturing face images via the phone-browser camera; created wireframes for the teacher dashboard.
To Build recognition result confirmation screen.	4 Oct 2026	31 Oct 2026	To be Built recognition-result interface ("Marked Present" confirmation screen); added error states for unrecognized faces; tested camera permissions across Chrome/Safari mobile browsers.
To Build attendance & profile screens for backend link	1 Nov 2026	2 Jan 2027	To be Built the attendance-history and student-profile screens ready ready for backend connection; implemented client-side form validation for enrollment.
To Build Mobile teacher dashboard & connect it to APIs	3 Jan 2027	6 Feb 2027	To be Built the Mobile-friendly teacher dashboard and connected it to backend APIs; ran across device responsive testing.
To Test UI across devices & lead final Presentation.	7 Feb 2027	6 Mar 2027	To be Tested the interface across multiple smartphones and low bandwidth conditions, improving offline-mode UI indicators with pending-sync badges; led final frontend testing during the pilot and delivered the final presentation.

Mentor's Name & Signature

Pawan Kr. Patidar

22/8/26